

Servlet & JSP

How to Perform JSP CRUD Application Using PostgreSQL ?

1. Create a PostgreSQL database and a table to store user data.
2. Design a JSP form to collect user input (name, email, etc.).
3. Use JDBC to connect JSP pages with the PostgreSQL database.
4. Write SQL queries to insert, display, update, and delete data.
5. Use PreparedStatement for executing queries securely.
6. Retrieve data using ResultSet and display it in a JSP table.
7. Redirect pages after each operation using response.sendRedirect().
8. Test all CRUD operations to ensure smooth database interaction.

How JSP Interacts with Databases Dynamically ?

1. **Database Connectivity:**
JSP uses **JDBC (Java Database Connectivity)** to establish a connection between the web application and the database.
2. **Driver Loading:**
The database driver (e.g., org.postgresql.Driver) is loaded using Class.forName() to enable communication with the specific database.
3. **Establishing Connection:**
A connection is created using DriverManager.getConnection(), which connects JSP to the database using the URL, username, and password.
4. **Executing SQL Queries:**
SQL queries like INSERT, SELECT, UPDATE, or DELETE are executed using **Statement** or **PreparedStatement** objects.
5. **Dynamic Data Handling:**
The JSP page retrieves user inputs with request.getParameter() and uses them in SQL queries to store or fetch data dynamically.
6. **Displaying Results:**
Data fetched from the database using a **ResultSet** is displayed on the web page in real-time using JSP expressions (<%= %>).
7. **Closing Connections:**
After execution, all database objects (ResultSet, Statement, Connection) are closed to release resources and maintain performance.

Modules of the Application :-

1. Registration Form (index.jsp)

- This page contains a simple **HTML form** that allows the user to input their details such as:
 - Full Name
 - Email
 - Password
 - Contact Number
- The form uses the POST method and submits the entered data to registration.jsp for processing.

2. Data Insertion (registration.jsp)

- This file retrieves the form data using the request.getParameter() method.
- It establishes a connection to the PostgreSQL database using the JDBC driver.
- A **PreparedStatement** is used to execute an INSERT query, which adds the user's data into the **student** table.
- Once the record is successfully inserted, the user is redirected to the **Welcome Page (welcome.jsp)** using response.sendRedirect().

Key Concepts Used:

- Class.forName("org.postgresql.Driver") – Loads PostgreSQL driver.
- DriverManager.getConnection() – Connects JSP to the PostgreSQL database.
- PreparedStatement – Used to safely insert data and prevent SQL Injection.

3. Display Records (welcome.jsp)

- This page retrieves all the stored records from the database using a SELECT * FROM student query.
- A **ResultSet** object is used to display data dynamically inside an HTML table.
- Each record is shown with a “Delete” link that passes the record ID to delete.jsp for deletion.

Purpose:

To view all registered users in a structured tabular format directly from the database.

4. Delete Operation (delete.jsp)

- This page performs the **DELETE** operation using the ID value passed from the welcome page.
- It prepares a SQL statement `DELETE FROM student WHERE id=?` to remove the selected record.
- After deleting, the page redirects back to the welcome page to refresh the record list.

Key Concept Used:

PreparedStatement ensures secure deletion of records without SQL injection risks.

Example – Registration Form

index.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <h1>Registration Form</h1>

    <form action="registration.jsp" method="post">
        <input type="text" name="fullname" placeholder="Enter your fullname"><br><br>
        <input type="email" name="email" placeholder="Enter your email"><br><br>
        <input type="password" name="password" placeholder="Enter your password"><br><br>
        <input type="number" name="contact" placeholder="Enter your contact number"><br><br>

        <button type="submit">Submit</button>
    </form>
</body>
</html>
```

registration.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%
        String fn=request.getParameter("fullname");
        String em=request.getParameter("email");
        String pss=request.getParameter("password");
        String cn=request.getParameter("contact");

        Class.forName("org.postgresql.Driver");

        String URL="jdbc:postgresql://localhost:5433/javabatch";
        String username="postgres";
        String password="4518";

        Connection con=DriverManager.getConnection(URL,username,password);

        String q = "insert into student(fullname,email,password,contact)values(?,?,?,?)";
```

```

        PreparedStatement ps = con.prepareStatement(q);
        ps.setString(1, fn);
        ps.setString(2, em);
        ps.setString(3, pss);
        ps.setString(4, cn);
        ps.executeUpdate();
        out.print("Registration Successfully Completed");

        response.sendRedirect("welcome.jsp");
    %>
</body>
</html>

```

welcome.jsp –

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <h1>Welcome Page</h1>

    <table border="1">
    <tr>
        <th>Id</th>
        <th>Fullname</th>
        <th>Email Id</th>
        <th>Password</th>
        <th>Contact</th>
        <th></th>
    </tr>

    <%
        Class.forName("org.postgresql.Driver");

        String URL="jdbc:postgresql://localhost:5433/javabatch";
        String username="postgres";
        String password="4518";

        Connection con=DriverManager.getConnection(URL,username,password);

        Statement stmt=con.createStatement();
        ResultSet rs=stmt.executeQuery("select * from student");
        while(rs.next()){
    %>
    <tr>
        <td><%= rs.getInt(1) %></td>
        <td><%= rs.getString(2) %></td>
        <td><%= rs.getString(3) %></td>
        <td><%= rs.getString(4) %></td>
        <td><%= rs.getString(5) %></td>
        <td>
            <a href="delete.jsp?id=<%= rs.getInt(1) %>" onclick="return confirm('Are you sure ?')">Delete</a>
        </td>
    </tr>
    <% } %>
    </table>
</body>
</html>

```

delete.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%
        int id=Integer.parseInt(request.getParameter("id"));

        Class.forName("org.postgresql.Driver");

        String URL="jdbc:postgresql://localhost:5433/javabatch";
        String username="postgres";
        String password="4518";

        Connection con=DriverManager.getConnection(URL,username,password);

        String q = "delete from student where id=?";
        PreparedStatement ps = con.prepareStatement(q);
        ps.setInt(1,id);
        ps.executeUpdate();

        response.sendRedirect("welcome.jsp");
    %>
</body>
</html>
```

Output –

Index.jsp Page -

← ↻ ⓘ localhost:8081/MyJSP_Project/index.jsp

Registration Form

welcome.jsp Page -

← ↻ ⓘ localhost:8081/MyJSP_Project/welcome.jsp

Welcome Page

Id	Fullname	Email Id	Password	Contact	
2	abc	abc@gmail.com	5489	8789654879	Delete
4	xyz	xyz@gmail.com	8564	9785649825	Delete

← ↻ ⓘ localhost:8081/MyJSP_Project/welcome.jsp

Welcome Page

Id	Fullname	Email Id	Password	Contact	
2	abc	abc@gmail.com	5489	8789654879	Delete
4	xyz	xyz@gmail.com	8564	9785649825	Delete

localhost:8081 says

Are you sure ?

← ↻ ⓘ localhost:8081/MyJSP_Project/welcome.jsp

Welcome Page

Id	Fullname	Email Id	Password	Contact	
4	xyz	xyz@gmail.com	8564	9785649825	Delete

Database - Student

	id [PK] integer 	fullname character varying (100) 	email character varying (100) 	password character varying (100) 	contact character varying (15) 
1	4	xyz	xyz@gmail.com	8564	9785649825